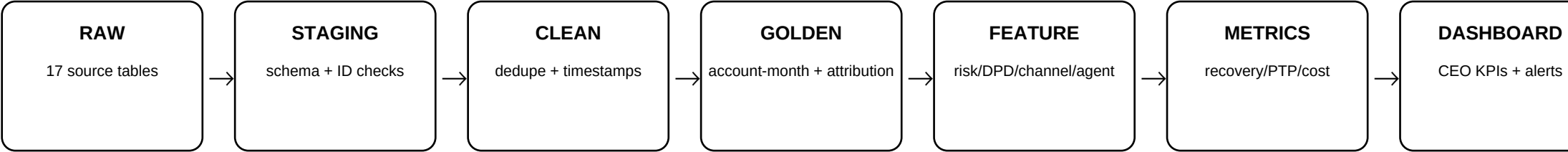


Production Analytics Architecture — Collections



Production controls

- Data contracts: keys, nullability, enums, timezone contract.
- Incremental processing: event-time watermark + late-arrival lookback; deterministic upserts.
- Backfills: rerun affected partitions with versioned metric snapshots.
- DQ monitoring: duplicates, orphan accounts, campaign-window validity, timezone mismatch, payment reconciliation.
- Anomaly detection: recovery/account, recovery/agent-hour, paid rate, kept PTP, complaints, vendor/channel shifts.

Metric principle

Executive KPIs are computed only from the Golden layer and carry definitions, lineage, and DQ status.